

Lecture 2 Tutorial — Data and Statistics

Q1. Key Terms (Fill in the blanks)

Complete the statements.

1. A parameter describes a _____.
2. A statistic describes a _____.
3. A sample is a subset of a _____.

Q2. True / False

Write T (True) or F (False).

1. A larger-looking bar in a chart always means a larger real difference.
2. A sample can be large and still be biased.
3. Correlation proves causation if the value of r is high.
4. Missing data can affect the validity of conclusions.

Q3. Multiple Choice (Sampling Bias)

A lecturer posts a survey link in the class group chat and only students who choose to respond are included. This is best described as:

- A. Stratified sampling
- B. Systematic sampling
- C. Voluntary response sampling
- D. Cluster sampling

Q4. Variable Type Classification

Classify each variable as Quantitative or Categorical. If quantitative, also classify it as Discrete or Continuous.

1. INT104 quiz score (0–20)
2. Number of AI-tool uses per week
3. Student major (AI / CS / DS)
4. Time spent on revision (hours)

Q5. Missing Data (Comparison Table)

Complete the table with one advantage and one risk for each method.

Method	One advantage	One risk
Delete cases		
Impute missing values		

Q6. Sampling Methods (Matching)

Match each description to the correct method.

Methods:

- A. Convenience sampling
- B. Systematic sampling
- C. Stratified sampling
- D. Cluster sampling

Descriptions:

- 1. Select every 8th student on a list
- 2. Divide students by year level and sample from each year
- 3. Survey students leaving the library entrance
- 4. Randomly choose 2 tutorial groups and survey all students in those groups

Q7. Experimental Design (Structured Short Answer)

A tutor wants to compare traditional revision and AI-assisted revision for an INT104 test.

Fill in the design plan by giving a brief phrase for each item:

- 1. **Randomness:** How will students be assigned?
- 2. **Control of confounders:** Name one confounder to control.
- 3. **Replication:** What does replication mean here?
- 4. **Outcome variable:** What will be measured?

Q8. Histogram Concepts (Short Answer)

Answer the following briefly.

- 1. What is a histogram used to show?
- 2. What can happen if the number of bins is too small?
- 3. What can happen if the number of bins is too large?

Q9. Distribution Shape (Fill in the blanks)

For a right-skewed distribution, complete the relationship using $>$, $<$, or $\approx$:

Mean____Median____Mode

Then state whether the right tail is longer or shorter than the left tail.

Q10. Correlation and Significance (Interpretation)

Given:

Sample correlation:

$$r = 0.591$$

Critical values (class example):

$$\pm 0.878$$

Answer:

- 1. Is the correlation statistically significant under this criterion? (Yes / No)
- 2. Complete the reason: It is significant only if r is _____ than or _____ than _____.

Q11. Boxplot Practice (Calculation + Interpretation)

A dataset has the following five-number summary:

$$\text{Minimum} = 10$$

$$Q_1 = 25$$

$$\text{Median} = 35$$

$$Q_3 = 50$$

$$\text{Maximum} = 110$$

A modified boxplot shows 105 and 110 as outliers; the highest non-outlier value is 75.

Answer:

1. Compute the IQR.
2. State the interval containing the middle 50% of observations.
3. Based on the high-end outliers, the distribution is more likely to be:
 - A. Left-skewed
 - B. Right-skewed
 - C. Symmetric